

Uveal Melanoma: Proton Beam Radiation Therapy

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Anne Marie Lane, Megan S. Steinkerchner,
Evangelos S. Gragoudas, and Ivana K. Kim

15.1 Introduction

Radiotherapy (RT) is the first choice of treatment for the vast majority of malignant melanomas of the uveal tract. Eye conservation and, in many cases, maintenance of useful vision are possible. Two major radiotherapeutic techniques for the treatment of uveal melanomas are available, brachytherapy with radioactive plaques, which are sutured on the sclera over the area of the tumor [1], and external beam irradiation using charged particles [2, 3]. Both plaque radiotherapy and external beam irradiation are effective for treating small, medium, and most large uveal melanomas; high rates of local control are achieved, and survival rates are similar to those observed after enucleation [4–6], which is reserved for very large tumors, tumors with large extrascleral extension, or patients with blind, painful eyes.

The physical properties of charged particles such as protons allow localized dose distributions [2, 7, 8]. Stereotactic radiosurgery does not appear to have as favorable dose distributions [9, 10]. Most published studies have reported results in small patient cohorts. Findings thus far indicate higher rates of complications and visual loss when compared to brachytherapy [11] or proton beam radiotherapy [12].

15.2 Proton Beam Radiotherapy

In 1946, Robert R. Wilson was the first to recognize that the physical properties of protons [13, 14] (Box 15.1) could be advantageous for medical purposes [15], including treating tumors. Today, proton radiation is the most widely used form

Box 15.1 Proton Beam Radiotherapy of Uveal Melanoma

- Delivery of maximum density of ionization as protons stop (Bragg peak).
- Sharp reduction of the dose outside the treated area.
- Modulation of beam energy: Bragg peak can be broadened to cover a tumor at any depth.

of external beam radiotherapy (EBRT) for the treatment of uveal melanoma.

Because of these properties, proton radiation is particularly advantageous for tumors that are large and/or located near the optic nerve or macula. Proton therapy in these cases can provide improved local control and reduced radiation-induced collateral damage to normal ocular tissues. Patients with large, anteriorly located tumors are able to maintain vision because the proton beam can enter the eye directly in the area of the tumor, avoiding radiation exposure to noninvolved ocular structures.

The availability of proton facilities has increased in recent years. However, in the United States, only a few of these centers treat patients with ocular melanoma. There are also facilities in Canada, England, Germany, Russia, Switzerland, France, and Italy. With this increased access to proton therapy, more patients with uveal melanoma can now consider proton irradiation as a treatment option. Newer cyclotrons are producing high-energy beams, which are less precise than currently used low-energy beams. As a result, healthy tissues around the tumor receive higher doses of radiation. Several groups are conducting research to mitigate this problem (e.g., using two-beam strategies [16, 17]).

15.3 Patient Evaluation

Prior to treatment, patients have a complete ophthalmological examination as well as fundus photography, A-scan and B-scan ultrasonography, and axial length measurement.

A. M. Lane (✉) · M. S. Steinkerchner · E. S. Gragoudas · I. K. Kim
Ocular Melanoma Center, Department of Ophthalmology,
Massachusetts Eye and Ear Infirmary/Harvard Medical School,
Boston, MA, USA
e-mail: Anne_Lane2@meei.harvard.edu;
msteinkerchner@meei.harvard.edu;
evangelos_gragoudas@meei.harvard.edu;
ivana_kim@meei.harvard.edu

Some centers also perform magnetic resonance imaging and/or computerized tomography [18]. Staging imaging is performed before treatment to rule out metastasis. Proton radiation is generally feasible to treat tumors that occupy up to 30% of the ocular volume.

15.3.1 Pre-radiation Surgery

During a surgical procedure, the tumor is localized by transillumination and indirect ophthalmoscopy, and the episcleral tissues over the tumor are examined for evidence of extra-scleral extension. The tumor margins are identified and marked on the sclera with a surgical pen. Four 2.5-mm radiopaque fiducial tantalum rings (also known as “clips” or “markers”) are sutured on the sclera around the borders of the tumor (Fig. 15.1), which serve as reference points for the placement of the proton beam at the time of treatment. For tumors with ciliary body or iris involvement, rings are placed at the posterior margin of the tumor, and the distance from the rings to the anterior margin of the lesion is measured. If tumors are in contact with the optic nerve, rings are placed only at the anterior and lateral margins of the lesion, and the distance from the rings to the posterior margin is estimated by indirect ophthalmoscopy and from fundus photographs. Advances in ocular imaging have stimulated interest in administering proton beam radiotherapy without having to insert fiducial rings [17], which would be helpful in patients

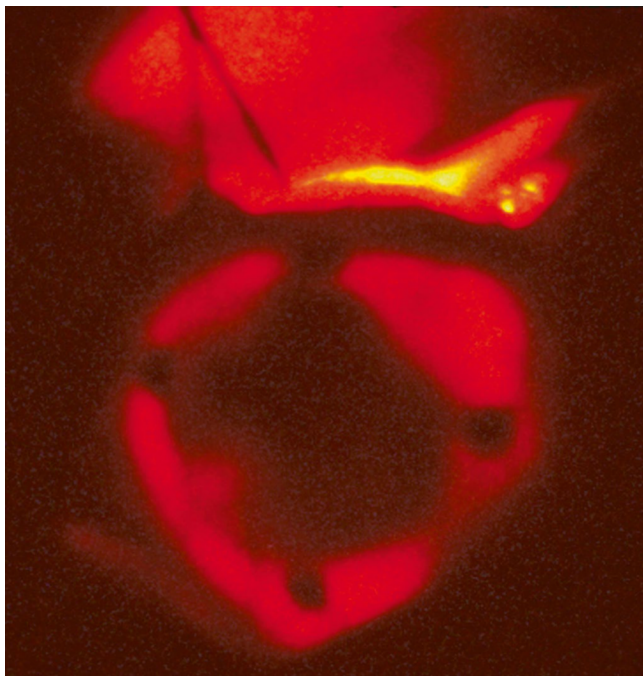


Fig. 15.1 Tantalum rings sutured to the sclera at the edges of the tumor seen by transillumination. (reprinted from [19]. With permission from Elsevier)

with poor general health and in those with thin sclera (e.g., high myopes).

After the rings are sutured, distances from each ring to tumor margin, to corneal limbus, and to every other ring are measured with calipers. Drawings are then made to document the size and shape of the lesion as well as the location of the tantalum rings. Trans-retinal or trans-scleral tumor biopsy (fine needle aspiration biopsy (FNAB)) for molecular prognostic testing may also be performed at the time of this surgery. Some genetic analyses, such as multiplex ligation-dependent probe amplification (MLPA) and next-generation sequencing (NGS) may be performed on tumor biopsies taken after completion of the radiotherapy [20], which may be done if there is concern for risk of extraocular tumor seeding, which is extremely rare [21].

Surgery is not necessary for patients with tumors involving only the ciliary body or iris because transillumination and/or photographs can be used to define tumor margins in relation to the anatomic landmarks of the iris and conjunctiva.

15.3.2 Proton Treatment Planning

The interactive treatment planning system developed for ocular melanomas [EYEPLAN, Martin Sheen, Clatterbridge Centre for Oncology, Bebington, UK] and used for all patients seen at our site has been previously described [2, 22]. A team of ocular oncologists, radiation oncologists, and physicists use axial eye length and tumor thickness, determined by A- and B-mode ultrasonography, tumor basal dimensions measured using ophthalmoscopy, transillumination, and ultrasound, and location of the tantalum rings, determined by orthogonal X-rays taken in the treatment position during a simulation session, to develop a specific treatment plan for each patient. As mentioned, some centers also incorporate MRI and/or CT imaging for treatment planning [18]. Newer planning systems, such as OCTOPUS and RayOcular are being developed to overcome the limitations of EYEPLAN [17, 23]. These data are used to create a three-dimensional model of the tumor that is superimposed on a generic model of a normal eye, scaled to the ultrasonically determined length of the patient's eye. There is a growing interest in replacing these generic eye models with real-eye anatomy for treatment planning, using MRI and CT imaging [16]. The fundus photographs are important when creating models for very posterior tumors or tumors abutting the optic nerve because these tumors are not fully surrounded with marker rings. Two tumors can be modeled when a tumor is shaped irregularly. Iris tumors and ciliary body tumors are drawn from clinical and ultrasound information. The computer can be made to rotate the eye so that it follows a user-controlled fixation point in near-to-real time, which allows the planner to choose a fixation angle that will minimize

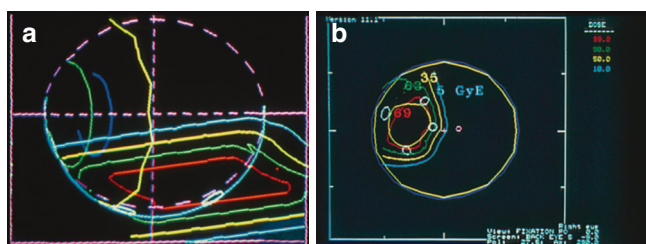


Fig. 15.2 Graphic depiction of dose distribution from treatment plan for 13-mm-diameter tumor. Isodose contours shown in a plane through the eye (a). Isodose contours shown on the retinal surface (b)

radiation exposure of the lens, optic disk, and fovea to the extent possible [22].

This program also automatically designs an aperture in a brass disk that approximates the shape of the tumor and gives a 3 mm margin (which can vary between 2.5 and 4 mm if needed) around the tumor at the 50% dose level. It calculates the maximum and minimum depths of the tumor as well as the beam range and modulation width, based on user-selected proximal and distal margins. A spinning propeller is used to create a number of Bragg peaks so that the dose-depth curve changes from a sharp peak to a plateau according to these margins. A wedge can also be inserted into the beam to taper the distal end of the beam (e.g., to reduce dose to optic nerve). Program-generated fundus dose distributions are displayed in the geometry of a wide-angle fundus photograph and in any plane through the eye (Fig. 15.2). Dose-volume histograms for the tumor and many structures of the eye (e.g., globe, lens, ciliary body, retina, macula, and disk) can be generated for each treatment plan.

15.4 Irradiation Procedure

Prior to the start of irradiation, a positioning procedure is performed during which the patient must be immobilized and the eye properly aligned with the proton beam. The patient is treated in the seated position, immobilizing the head by using an individually contoured face mask and bite block, which are attached to a head holder. The head holder is attached to the proton beam collimator, and the aperture is mounted in the end of the collimator nearest to the eye (Fig. 15.3). Orientation of the patient's eye is established by voluntary fixation of the eye on a flashing light set at a position determined by the planning program. If the vision is poor in the eye to be treated, the other eye, which is usually covered, can be used for fixation.

Using a fluoroscopic system that provides a virtually instantaneous picture held on an image-storage device, the alignment of the tumor relative to the beam axis and the edges of the beam-defining aperture is achieved by moving the head holder until the surgically placed tantalum rings that

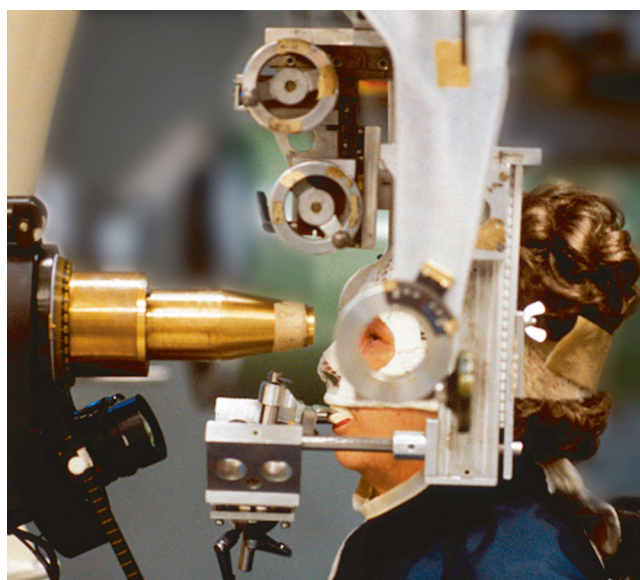


Fig. 15.3 Patient immobilized for treatment with the use of head holder, face mask, and bite block. (reprinted from [19])

spatially define the tumor are in the desired position relative to the beam axis. Finally, a beam simulation field light is projected through the aperture onto the eye to be treated, to ensure that the light field falls on the eye in relation to the edge of the limbus. For tumors treated without surgical localization, a light beam coaxial with the central axis of the proton beam is used to position the tumor relative to the beam during treatment [24, 25]. The alignment procedure lasts approximately 15 minutes. After successful completion of this step, treatment can begin.

During treatment, the eyelids are retracted by the ophthalmologist using a lid speculum to reduce radiation exposure to the eyelids, and the patient is asked to look at the fixation target. Using the same fluoroscopic system used for patient positioning, a magnified picture of the eye, taken by a camera mounted on the collimator, is viewed onscreen in the control area to monitor the eye position. This system allows the interruption of treatment and repositioning of the patient if unacceptable eye movement (more than 0.5 mm) is observed. Each treatment takes approximately 1 minute, and almost all treatments are completed without interruption.

The standard total dose is 70 Gy relative biologic effectiveness (RBE) (63.6 proton Gy times 1.1 relative biologic effectiveness (RBE)) administered in daily fractions of 14 Gy (RBE) over 5 days. We chose this fractionation scheme based on favorable clinical results demonstrated with the use of a small number of relatively large dose fractions in patients with skin melanomas [7, 26]. At the prescribed dose of 70 Gy (RBE), we estimate that the optic nerve and macula receive the full dose when the tumor is less than 1 mm from these structures and half the dose (35 Gy (RBE)) when the tumor is located 3 mm from these structures.

The maximum tolerated and minimum effective doses of proton radiation have yet to be established. However, the findings of a dose reduction trial [27] indicated that treatment with 50 Gy (RBE) did not compromise tumor control in patients with small- or medium-sized tumors located near the optic nerve or fovea. As a result, in this select group of patients, we administer a dose of 50 Gy (RBE) to potentially reduce vision-threatening radiation-induced complications. There was no benefit found in terms of visual acuity in a more recent evaluation of patients with paramacular tumors treated with the 50 Gy (RBE) dose [28]. At several facilities in Europe, a total dose of 60 Gy (RBE) is given in four equal fractions over 4 days [29–31].

Proton beam radiotherapy for iris melanomas is performed without the use of fiducial rings, using the beam simulation field light to guide positioning [25, 32–34]. Diffuse iris melanomas are treated with whole anterior-segment proton beam irradiation [35, 36] although there is an increased risk of ocular morbidity with such extensive treatment [37].

15.5 Follow-Up Protocol

After treatment, ophthalmological examinations are performed 6 weeks, then every 6 months up to 5 years, and then annually to monitor the tumor and patient status and to detect any side effects and complications. Fundus photography and ultrasonography are performed at varying intervals to document tumor regression, progression or relapse, and any radiation-induced morbidity. Metastasis surveillance includes liver imaging and is now tailored to molecular prognostic information obtained through tumor biopsy if available.

15.6 Patient Characteristics

The largest series of uveal melanoma patients has been treated at the Massachusetts Eye and Ear (MEE)/MGH Francis H. Burr Proton Therapy Center. Over 5000 patients have been treated beginning in 1975 at MEE, and long-term follow-up data continue to accrue.

In a prior report from MEE analyzing 2069 patients treated at the Harvard Cyclotron between 1975 and 1997, approximately equal numbers of men and women were treated, and the mean age of these patients was 61 years. The median basal diameter and median tumor height of the treated tumors were 15.2 mm and 5.3 mm, respectively, and 68% of these were located within 3 mm of the optic nerve or fovea [38].

Patients with comparable demographic and tumor characteristics have been treated at other proton facilities following essentially the same or a somewhat modified planning protocol as that developed at the Harvard Cyclotron [29–31].

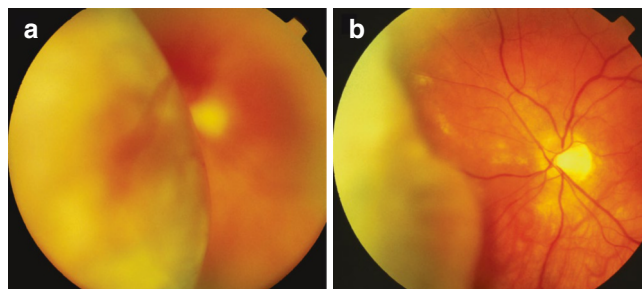


Fig. 15.4 Large ciliochoroidal melanoma extending up to the optic nerve and associated with serous retinal detachment. Before proton irradiation (a). Approximately 1 year posttreatment, significant reduction of the tumor is seen (b)

15.7 Ophthalmic Outcomes

15.7.1 Local Control

Disappearance of the tumor or formation of a flat scar occurs infrequently, and the vast majority of lesions continue to regress years after therapy. Regression (Fig. 15.4) is most likely due to both direct, senescent cell death at the time of mitosis, radiation damage to chromosomal DNA, and tumor ischemia from damage to its vasculature. Delayed regression in some patients may be due to the prolonged intermitotic phases (i.e., doubling time) of melanoma cells.

Local recurrence after charged particle irradiation is observed in 2–5% of patients [29, 31, 39]. In the MEE series [38], approximately 3% of tumors exhibited growth, confirmed by ultrasonography and fundus photography, and just under one-half of these were marginal recurrences. The highest annual rate of recurrence, 1%, occurred 1 year after proton therapy, with the earliest occurrence observed 5 months after irradiation and the latest after 11 years. The 15-year probability of local tumor control, based on confirmed and possible recurrences combined ($N = 60$), was 95% (95% CI, 93–96%). Similar control rates were reported in a series of patients treated at Center Hôpital Ophtalmique Jules-Gonin/Paul Scherrer Institute (PSI) [29]. In some cases, eyes with local recurrence can be retreated successfully with repeat proton irradiation [40] or—for cases in which the recurrence is marginal and flat—transpupillary thermotherapy.

Higher recurrence rates have been observed after brachytherapy, probably because of inaccurate plaque positioning over tumors located far posteriorly [41]. Proton irradiation may be a better choice of treatment for such tumors, and a recent study of long-term outcomes after proton irradiation for patients with tumors extending within 1 disk diameter (DD) of the optic disk supports this view. The 5-year tumor recurrence rate with such tumors was approximately 3%, virtually identical to the rate seen with tumors farther away from the nerve during this period [42]. In contrast, after

brachytherapy, 10% of tumors overhanging the disk recurred [43]. Advances in plaque design may improve tumor control in these cases [43–45].

Inability to achieve local control is associated with an increased risk of metastasis and poorer survival [29, 31, 46]. In a retrospective study of 4196 patients treated with proton beam irradiation (PBI) at MEE, the 10-year rate of melanoma-related mortality was 15% in patients without local recurrence and 46.6% in patients with recurrences [47]. The Ophthalmic Oncology Task Force reported a 10-year metastasis-free survival rate of 82% for patients without recurrences and 62% for those with local recurrences. Multivariate analysis adjusting for established prognostic factors such as AJCC stage, ciliary body involvement, or extrascleral extension continued to illustrate local recurrence was associated with increased risk of metastasis [48]. Whether the recurrence causes metastatic disease or whether it is only a marker of a particularly aggressive tumor type is still uncertain.

15.7.2 Eye Loss

The probability of retaining the eye for patients treated at the MEE was 91% at 5 years, 88% at 10 years, and 84% at 15 years after irradiation, and the rates at these time points were virtually identical in patients treated at PSI [49].

Removal of the eye may become necessary after treatment if the tumor recurs or complications develop. Neovascular glaucoma is the most common complication leading to enucleation [38, 46]. Tumor characteristics that increase the risk of enucleation include proximity of the tumor to the optic nerve [38, 49, 50], and greater tumor size (tumor height [30, 31, 46, 49] and tumor diameter [30, 31, 38]). In an analysis of patients with large tumors (i.e., with height ≥ 10 mm or diameter > 16 mm, or height exceeding

8 mm when the optic nerve was involved) treated at MEE, enucleation was necessary in approximately 23% of patients 5 years after treatment, and this increased to almost 30% at 10 years after PBI. Five-year rates of enucleation were higher (31%) for tumors located near the optic nerve and fovea and lower (12%) for tumors located farther away from both structures [51]. In another study of patients with large tumors, defined as T3–T4 tumors, 19.5% of the cohort underwent enucleation [52].

Eyes enucleated after proton therapy and examined histopathologically exhibit degenerative and vascular changes [53]. Thickening or thrombosis of the tumor vasculature is a hallmark effect of radiation; in one study, vascular damage was identified over ten times more often in irradiated tumors than nonirradiated tumors [53]. Consistent with these findings, fewer vascular regions were identified by color Doppler imaging in proton-irradiated tumors versus pre-irradiation tumors [54].

15.7.3 Vision Loss

Studies at MEE have shown that risk factors for vision loss to 20/200 or worse after proton beam irradiation depends on several factors, which include greater tumor height, greater tumor diameter, tumor margin less than 3 mm from the optic nerve and/or fovea, initial visual acuity less than 20/40, documented tumor growth, extensive retinal detachment, patient age greater than 60 years, and history of diabetes mellitus [28, 38, 51, 55, 56]. Based on risk scores derived using these factors, the 10-year probabilities of visual deterioration (20/200 or worse) were 16% for patients in the “low-risk” category and 99% for patients in the “high-risk” group (Table 15.1).

Our findings have been supported by Fuss et al. in London [50] and by Damato et al. in Liverpool [30].

Table 15.1 Cumulative rates (percent and 95% confidence intervals) of enucleation, vision loss, and melanoma-related death, by risk group^a and years after proton irradiation

Endpoint	All patients	Low risk	Low-to-medium risk	Medium-to-high risk	High risk
Loss of eye					
5 years	9 (7, 10)	2 (1, 3)	6 (4, 8)	10 (7, 14)	27 (22, 33)
10 years	12 (10, 14)	3 (2, 6)	8 (6, 11)	14 (10, 20)	33 (27, 41)
15 years	16 (13, 20)	3 (2, 6)	11 (7, 16)	23 (15, 35)	48 (33, 65)
Vision loss to 20/200 or worse					
5 years	52 (50, 55)	9 (7, 15)	40 (35, 45)	71 (67, 76)	95 (91, 98)
10 years	65 (62, 68)	16 (11, 22)	58 (52, 65)	84 (79, 88)	99 (96, 100)
15 years	71 (66, 75)	24 (14, 39)	65 (55, 75)	88 (81, 93)	100 (–, –)
Melanoma-related death					
5 years	14 (13, 16)	2 (1, 4)	7 (5, 9)	19 (16, 22)	40 (35, 47)
10 years	23 (21, 25)	3 (2, 5)	13 (11, 17)	31 (27, 35)	57 (51, 64)
15 years	27 (24, 29)	5 (3, 9)	16 (13, 19)	35 (31, 40)	63 (56, 71)

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^aRisk groups derived from values of statistically significant prognostic factors and their coefficients in multivariate Cox regression models

15.7.4 Radiation-Induced Ocular Morbidity

The most serious problem is neovascular glaucoma, which increases the risk of vision loss and loss of the eye. In a series of patients with tumors too large to be treated with plaque radiotherapy, larger tumor diameter and retinal detachment were highly significant risk factors ($p < 0.0005$) for developing rubeosis [58]. In the MEE cohort, the most significant predictor of iris neovascularization was larger tumor volume, with a relative risk of 2.4 (95% CI: 2.1, 2.8) per doubling volume. Annual rates of developing one or both complications were highest in the first 3 years after proton beam radiation and dropped to 1% or less 5 years or longer after treatment [57].

Damato introduced the concept of “toxic tumor syndrome,” which is caused by radiation-induced, intra-tumoral vasculopathy resulting in (a) vascular occlusion causing tumor ischemia, and release of angiogenic factors, causing rubeosis and neovascular glaucoma, and (b) vascular incompetence leading to macular edema, lipid retinopathy, and serous retinal detachment (Chap. 18) [37]. Methods of treating toxic tumor syndrome, neovascular glaucoma, radiation vasculopathy, and other radiation-induced vascular complications include intravitreal injections of antiangiogenic agents (e.g., bevacizumab, infliximab) [59–63] and/or steroids [64, 65], transpupillary thermotherapy applied to the tumor, and tumor removal by endoresection or eyewall resection. Studies investigating the scope of prophylactic antiangiogenic therapy are producing encouraging results, but the most efficient protocol has yet to be defined.

Studies at MEE report overall 5-year cumulative rates of maculopathy and papillopathy of 40% and 24%, respectively [66]; however, these rates increase to 64% and 35%, respectively, in patients with small- to moderate-sized tumors extending to within 4 disk diameters of the optic nerve or fovea [66]. In an analysis of patients with tumors located within 1 disk diameter of the optic nerve [42], 57% had developed radiation papillopathy with vision worse than 20/200 in 80% by 5 years after treatment. By 10 years after treatment, there was little additional change in papillopathy rates (61%), but vision loss was observed in 91% of patients. In another study that limited analysis to parapapillary melanomas located within 1 disk diameter of the optic nerve and at least 2 disk diameters from the fovea, the 5-year rate of papillopathy was 81% in patients with tumors touching the optic nerve compared to 67% in patients with tumors located 0.1–1 disk diameter from the optic nerve. Total loss of vision was not an inevitable consequence of developing papillopathy; 42% of patients retained vision of counting fingers or better, and 31% ($n = 13$) experienced recovery of visual acuity (i.e., gain of two or more lines) after an initial loss [67]. In these analyses, this morbidity was related to tumor distance from these structures and, therefore, total radiation

dose to them. In addition, many tumors in these studies were large and also involved the fovea.

A randomized clinical trial to establish safety and efficacy of reducing the dose from 70 to 50 Gy (RBE) was completed in eyes with tumors extending to within 4 DD of optic nerve and/or fovea but not exceeding 5 mm in height and/or 15 mm in diameter. The lower dose reduced loss of visual field but not loss of visual acuity [27]. However, local control and mortality were not adversely affected by the dose reduction. As a result, patients with small- or medium-sized tumors located within 1 DD of the optic nerve or fovea are now treated with the lower dose of 50 Gy (RBE). Similar results were observed in a more recent analysis of a larger cohort, comparing 388 patients who received lower-dose PBI to 732 patients who received the standard dose [68]. Other strategies must be found to improve functional outcomes after proton therapy. Hyperfractionation of the dose may reduce the risk of collateral damage to critical structures. Prevention and treatment of radiation-induced vascular damage may reduce the incidence and severity of maculopathy.

Although visual prognosis is generally poor for tumors located close to the optic disk or fovea, proton irradiation of these tumors should be considered because plaque radiotherapy may be an unsuitable treatment option. Visual acuity of counting fingers or better can be maintained in over half the cases up to 5 years after proton therapy (Fig. 15.5).

In a study of patients with iris melanomas conducted in Charité–University Hospital, Berlin, 60% developed secondary glaucoma 5 years after PBI treatment, and the strongest risk factor was the size of treatment fields [34]. The cause of glaucoma was not specified but is likely to be outflow obstruction by the tumor, especially when circumferential spread was more extensive as indicated by the size of the treatment fields. If there is confidence that the tumor has been sterilized by the proton beam radiotherapy, it may be possible to treat this glaucoma with Baerveldt shunts [70].

Whole-anterior-segment proton beam radiotherapy for extensive, diffuse iris tumors can cause keratopathy and limbal stem cell failure, which can improve after 6–18 months so that only lubricants are required, although anti-inflammatory

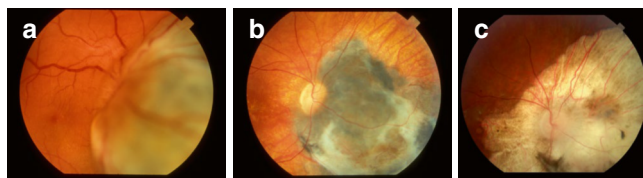


Fig. 15.5 Patient with a parapapillary melanoma touching the optic disk. Before proton irradiation, visual acuity was 20/40 (a). Two and a half years after treatment, tumor regression is seen but visual acuity is decreased (20/125) (b). Twenty-four years after proton irradiation, the tumor has regressed (<1.0 mm), and visual acuity is stable (20/100) (c). (Reprinted from [69]. With permission from Elsevier)

drops, bandage contact lens, amniotic membrane graft, or limbal stem cell transplantation may be necessary if the disease is severe or persistent [37].

Posterior subcapsular cataract (PSC) is associated with tumor height and radiation dose received by the lens, which in turn is correlated with the extent of the lens included in the treatment field [71]. The cumulative 5-year rate of PSC was 19% when less than 10% of the lens was included in the radiation field and 54% when more than 50% of the lens was irradiated [57]. Similarly, a study conducted with 107 iris melanoma patients in France and Italy treated with proton beam irradiation found that 57.4% of patients without cataracts before radiation developed cataract [72]. Cataract extraction can be performed without increasing the risk of local recurrence or metastasis and may offer some improvement in visual acuity in select cases [73, 74].

If the superior eyelid margin is not fully retracted out of the proton beam, keratinization of the superior palpebral conjunctiva can develop, causing painful corneal abrasion and, rarely, corneal perforation. This tends to occur with superior uveal tumors and can be prevented by transpalpebral proton beam radiotherapy, administered through the closed or half-closed upper eyelid with the eye looking downward so that the upper eyelid margin receives little or no radiation [75].

Tantalum rings can erode through irradiated conjunctiva if sutured anteriorly (i.e., less than 10 mm from limbus and within the radiation field). This results in scleral exposure and necrosis if the ring is not removed before conjunctival ulceration occurs.

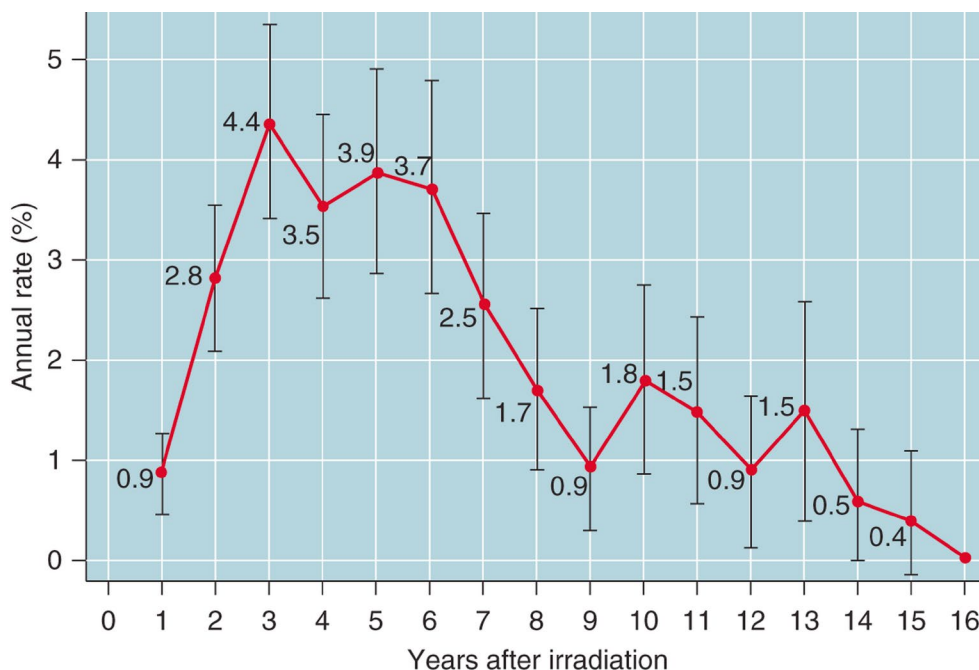
Proton beam radiotherapy for medial uveal tumors may unavoidably cause collateral damage to the lacrimal drainage system, with troublesome epiphora. Silicone lacrimal intubation gives poor results; conjunctivodacryocystorhinostomy with a Jones tube may be more successful [76].

Proton beam irradiation of the meibomian glands and the acinar cells of the lacrimal gland can result in meibum deficiency and aqueous deficiency dry eyes, respectively. Skillful detection and management of this morbidity are therefore important (Damato B, *Clinical Ophthalmic Oncology*, fourth edition, Volume 4, Chapter 15).

15.8 Metastasis and Survival

Five-, ten-, and fifteen-year rates of melanoma-related mortality were 14%, 23%, and 27%, respectively (Table 15.1), for patients treated with proton therapy at MEE. The highest annual death rates were observed between 3 and 6 years after treatment (Fig. 15.6). Patient and tumor characteristics associated with increased risk of metastatic death included larger tumor diameter, greater patient age, tumor pigmentation, presence of symptoms, ciliary body involvement, and green-blue iris color [38]. Investigators at PSI identified largest tumor diameter, patient age, tumor height, extrascleral extension of the tumor, degree of retinal detachment, and ocular melanocytosis as risk factors for metastatic death [29]. They also demonstrated that local failure was associated with a higher 10-year rate of melanoma-related mortality (i.e., 52.5% vs 27.4%, respectively) [29]. As mentioned previ-

Fig. 15.6 Annual rates of melanoma-related deaths, with 95% confidence intervals, up to 15 years after irradiation. (Reprinted from [57]. With permission from Elsevier)



ously, a MEE study found that 10-year rates of melanoma-related mortality were 15% and 46.6% without and with tumor recurrence, respectively [47].

Other reports of survival outcomes after proton irradiation originate from investigators at newer proton facilities and thus include fewer patients with limited follow-up. Survival rates at approximately 5 years after proton therapy are similar [30, 31] or somewhat lower [50, 77, 78] than those observed at MEE and PSI.

In previous decades, therapies for metastases from uveal melanoma have been largely ineffective, with a minority of patients surviving beyond 1 year of diagnosis (Chap. 28). However, recent advances may contribute to improved survival outcomes. Tebentafusp is the first drug approved by the FDA and EMA for metastatic uveal melanoma, with a randomized clinical trial demonstrating 1-year and 3-year overall survival (OS) rates of 72% and 27% in patients treated with tebentafusp versus 60% and 18% in patients treated with investigator's choice (pembrolizumab, ipilimumab, or dacarbazine) [79]. In a retrospective multicenter study evaluating real-world clinical experience with tebentafusp for metastatic uveal melanoma, the entire patient cohort experienced a median overall survival of 22 months. Patients treated with tebentafusp after prior treatment with immune checkpoint inhibitors (ICI) demonstrated an OS of 28 months as compared to 24 months for those receiving tebentafusp followed by ICI [80]. Additionally, the first liver-directed therapy indicated for hepatic dominant metastatic uveal melanoma approved by the FDA allows for local delivery of a high melphalan dose within the liver using a hepatic delivery system designed by Delcath. The FOCUS phase 3 trial found an OS of 20.5 months among patients treated with percutaneous hepatic perfusion of melphalan and 14.1 months among patients treated with best alternative care [81, 82].

15.9 Conclusions

Proton radiation plays an important therapeutic role for patients with uveal malignant melanoma. High rates of local control and eye conservation are achieved, with retention of useful vision in many cases. Proton irradiation accurately delivers a high homogeneous dose to the target tissue while allowing treatment of large tumors and tumors near the optic nerve and/or fovea. Nevertheless significant ocular morbidity can develop as a result of collateral damage to these structures and/or toxic tumor syndrome in large tumors. Anti-inflammatory and antiangiogenic agents have been shown to mitigate these deleterious effects. The Diabetic Retinopathy Clinical Research Network protocol AL will evaluate the prophylactic use of intravitreal faricimab and a fluocinolone acetonide implant for radiation retinopathy.

Although local control of the tumor is achieved in almost all cases, rates of melanoma-related mortality are still relatively high, particularly for patients with certain clinical and molecular tumor characteristics. With recent breakthroughs in treatments for metastatic disease and the ongoing search for effective adjuvant therapies, improved survival prognosis for future patients appears likely.

We can expect major advances in proton beam radiotherapy of uveal melanoma, which include improvements in ocular imaging, proton beam planning, dose optimization, and radiation delivery systems, as well as detection and treatment of radiation-induced morbidity.

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